Measuring fitness costs of evolved microbial populations

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Summary

Microbial populations are challenged by biotic (e.g predators and competitors) and abiotic (e.g., antibiotics and resource limitations) environments. Understanding how populations respond to these challenges via, for instance, standing genetic variation or *de novo* mutations is of great importance. Adaptations to novel environments come with fitness costs relative to the ancestral populations, which might be, for instance, a disadvantage in different environmental conditions.

Methods

Relative fitness

We have clones of E. coli strain B, E. coli REL607 (with a genomic marker Ara+) evolved in different environments, i.e., different antibiotic concentrations or in the presence or absence of lytic bacteriophage T4 stored at -80. We also have an ancestral clone stored at -80 °C. This is E. coli Strain REL606 with an isogenic Ara- genomic marker. Ara− and Ara+ cells produce red and white colonies, respectively, on Tetrazolium Arabinose (TA) agar plates, which help to differentiate ancestral and evolved lines on plates. We will recover the clones and compete ancestral and evolved lines together for 24 h to calculate fitness cost as follows:

1. In order to determine if any of the resistance has a fitness cost, all isolated clones of the evolved focal species will be replicated into 30 ml bottles containing 6 ml of LB medium and grown for 24 h at 30°C. The ancestral clone will also be grown in the same conditions.

2. Every clone will be diluted 100-fold (~10^7 cells ml^{-1}) and mixed 1:1 (v/v) with the ancestral strain.

3. Initial mixtures and after the 24 h growth at 30°C, cultures will be plated on TA agar with dilutions to define the relative abundances of ancestral and evolved clones using the following equation:

$$s = \frac{\ln\left[\frac{x_E(t)}{x_A(t)}\right] - \ln\left[\frac{x_E(0)}{x_A(0)}\right]}{t}$$

s… selection coefficient (>1 indicates higher fitness of evolved lines relative to the ancestor)

t … time interval (1 day)

$x_E(0)$ … number of colonies in the beginning of the evolved strain (*E.coli* B REL 607)

$x_A(0)$ … number of colonies in the beginning of the ancestral strain (*E.coli* B REL 606)

$x_A(t)$ … number of colonies after 1 day of the evolved strain (*E.coli* B REL 607)

$x_A(t)$ … number of colonies after 1 day of the ancestral strain (*E.coli* B REL 606)

Absolute fitness

Growth rates will be used to estimate focal species' absolute fitness across the microcosms. 1% of the recovered clones will be transferred into fresh LB media, and the OD_{600nm} will be measured at 15-minute intervals in an automated microplate at 30°C for 48 h. The measurements will be used to fit a logistic growth model. $N_t = KN_0e^{rt}/K + N_0(e^{rt} - 1)$ Where N_t is the observed population size at time *t.* Estimated parameters are: initial population size N_0, carrying capacity *K,* and growth rate *r.*

Phage resistance

Phage resistance will be estimated using the growth curves of clones mixed with phages. 1% of the recovered clones will be transferred into fresh LB media and supplemented with 10^7 phage particles, and the OD_{600nm} will be measured at 15-minute intervals in an automated microplate at 30°C for 48 h. Additionally, the virulence of the bacteriophage T4 was determined using the growth curves in the presence of T4 bacteriophages. Therefore, were added to the 3 replicates of diluted cultures. Resistance will be estimated using a liquid assay score by calculating the area under the curve for both growth curves with and without phages:

$$area\ under\ the\ curve = \sum_{i=1}^{24} \frac{OD_{i+1} + OD_i}{2}$$

$$liquid\ assay\ score = \frac{Area_{positiv\ control} - Area_{phage\ treatment}}{Area_{positiv\ control}} \cdot 100$$